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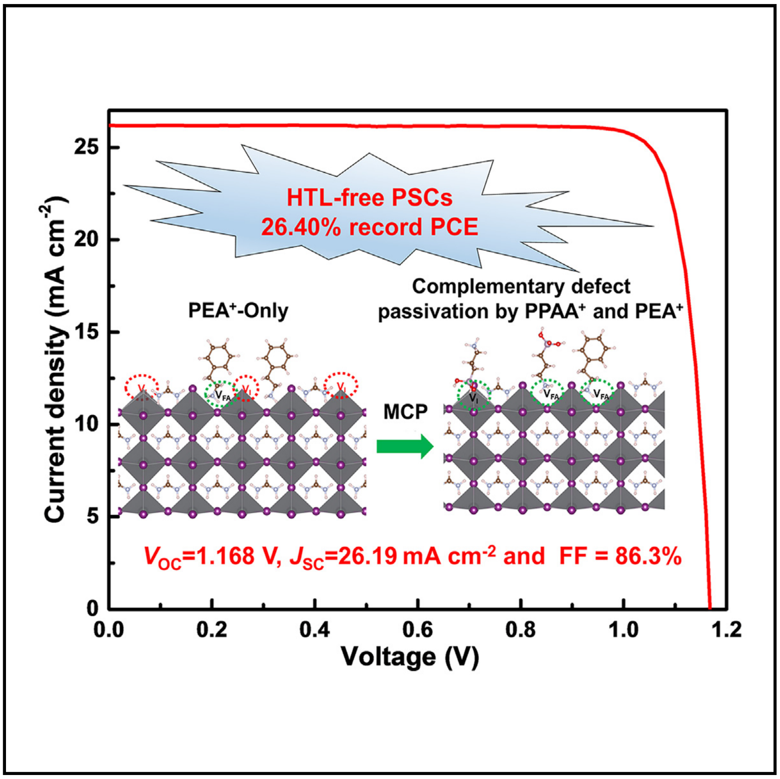
Enhancing electron transport for efficiency -recorded HTL-free inverted perovskite solar cells by molecular complementary passivation

Graphical abstract



Highlights

- Passivating synergically both V_I and V_{FA} defect states at the perovskite film surface
- Pushing surface Fermi level of perovskite closer to that of the electron-transport layer
- Advancing the efficiency record of HTL-free PSCs to 26.40% (25.92% certified)

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In brief

We report a novel molecular complementary passivation strategy to effectively passivate both typical perovskite surface defects of I and FA vacancies by combining propylphosphonic acid 3-ammonium bromide salt molecules with phenethylammonium bromide. This also pushes the perovskite surface Fermi level closer to that of the electron-transport layer and thus enhances interfacial electron extraction. A record efficiency of 26.40% (25.92% certified) was achieved for inverted perovskite solar cells (PSCs) without a predeposited hole-transport layer, also called p-n heterojunction PSCs.

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Enhancing electron transport for efficiency -recorded HTL-free inverted perovskite solar cells by molecular complementary passivation

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CONTEXT & SCALE Perovskite solar cells without pre-depositing a layer of hole-transport materials (HTL-free PSCs) are one of the most promising device structures in future PSC commercialization owing to their huge cost-down potential by simplifying the device structure and thereby reducing the use of materials and the related process. Depending on the molecule-extrusion process, the grain boundary and buried surface of the perovskite bulk film have been well managed. Further advancement of the device performance of HTL-free PSC needs critical interface engineering at the perovskite/electron-transport layer interface, where non-irradiative recombination occurs intensively due to the easily formed deep-level interfacial defect states (iodine vacancy [V_I] and FA vacancy [V_{FA}]). Here, we report a novel molecular complementary passivation (MCP) strategy with propylphosphonic acid 3-ammonium bromide (PPAABr) to mutually passivate surface defects of V_I and V_{FA} effectively. This results in a record efficiency of 26.40% (25.92% certified) for HTL-free PSCs and a competitive efficiency of 23.66% for 1.68 eV wide-band-gap PSCs, evidencing the generality of the MCP strategy.

SUMMARY

Inverted perovskite solar cells without pre-depositing a layer of hole-transport materials (HTL-free PSCs) still suffer from the non-irradiative recombination at the perovskite/electron-transport layer (ETL) interface. In this work, we report a molecular complementary passivation (MCP) strategy by employing propylphosphonic acid 3-ammonium bromide (PPAABr) to cooperate with phenethylammonium bromide (PEABr) to mutually passivate surface defects of I and formamidinium (FA) vacancies by multi-coordination. This passivation led to an obvious decrease in interfacial defect-state density and greatly improved exciton and carrier lifetime for the perovskite film. Moreover, MCP surface treatment pushes the perovskite surface Fermi level closer to that of ETL, thereby enhancing interfacial electron extraction. As a result, MCP-based HTL-free PSC achieved a record efficiency of 26.40% (25.92% certified). The encapsulated device retains 94.8% of its initial efficiency after 1,000 h of light soaking. The generality of the MCP strategy also generated a competitive efficiency of 23.66% for 1.68 eV wide-band-gap PSCs.

INTRODUCTION

Single-junction perovskite solar cells (PSCs) have experienced impressive progress, with the certified power conversion efficiency (PCE) exceeding 26%,[1–3] announcing their huge potential for commercialization. Without pre-depositing a layer of hole-transport materials (HTL), HTL-free PSCs have attracted increasing attention because they avoid the degradation of HTL materials and the interfacial decomposition of perovskites, more than leaving out the cost of materials and related processes.

In 2014, Hu et al. first reported a 5.4% HTL-free inverted PSC with an $MAPbI_3$ absorber film prepared by a sequential vapor deposition process.[4] In 2015, Tsai et al. adopted the antisolvent quenching method to obtain a higher quality $MAPbI_3$ film and achieved a device PCE of 11.02% in a configuration of indium-tin-oxide (ITO)/$MAPbI_3$/phenyl-C61-butyric acid methyl ester ($PC_{61}BM$)/Bis-C_{60}/Ag,[5] which was pushed to 16% by further improving the perovskite film quality.[6] Later on, Ye et al. innovatively incorporated the p-type cuprous thiocyanate (CuSCN) into the perovskite precursor solution to deposit a CuSCN-doped perovskite film and form a p-n heterojunction with a stronger built-in field for effective e-h dissociation, achieving a PCE of 18.1%.[7] Since then, more p-type molecules have been exploited to realize p-doped perovskite film and enhance carrier separation and extraction at the ITO/perovskite interface by modulating the electronic structure of the doped perovskite film, including CuI,[8] Cu(thiourea)I,[8] 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane,[9] and Cu(thiourea)Cl.[10] The PCE of the HTL-free device was promoted to 22.2%.[10]

In 2023, Zheng et al. innovatively introduced a series of self-assembling molecules (SAMs) used popularly, such as [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) and [2-(9H-carbazol-9-yl) ethyl]phosphonic acid (2PACz), into the perovskite precursor solution and advanced its PCE to 24.5%.[11] Meanwhile, our group designed a new SAM molecule, (4-(2,7-dibromo-9,9-dimethylacridin-10(9H)-yl)butyl)phosphonic acid (DMAcPA), to push the PCE record to 25.86% and discovered the clear distribution of DMAcPA in the grain boundary and on the buried surface of the perovskite layer, which favors constructing a well-matched p-perovskite/ITO Schottky contact and thus broke the record of single-junction PSCs.[12] This provides a reliable guideline of perovskite doping and Schottky contact construction for the development of HTL-free PSC.[12] Thereafter, Wu et al. successfully introduced 2PACz into tin-lead mixed perovskite to obtain a much-improved device PCE and stability.[13] Most recently, the Wu group further elevated the performance of HTL-free tin-lead mixed PSCs by realizing n-doped perovskite to enhance interfacial electron transport at the cathode contact.[14] Although the perovskite film buried surface has been well passivated and organized, the current HTL-free inverted PSCs lack the pungent upper-interface management of the perovskite film in the device, where non-irradiative recombination occurs intensively due to easily formed deep-level interfacial defects. Beyond any question, this hinders the advancement of HTL-free devices. Meanwhile, fortunately, the management of perovskite/electron-transport layer (ETL) interface has attracted increasing attention and proved to be encouraging in enhancing interfacial electron transport and suppressing non-irradiative recombination.[15–18] Depending on hydrogen bonding or Lewis acid-base coordination interactions with perovskite,[19,20] however, reliance on a single molecule effect may fail to address simultaneously the passivation issue of the complex chemical environment at the interface between perovskite and C_{60} or its derivatives.[17,21] On the other hand, the incorporation of extensively utilized large ammonium cations may not yield a continuous two-dimensional (2D) perovskite layer at the surface of the perovskite bulk film,[22,23] leaving some residual defect sites, especially exposed leads at the surface. Therefore, molecular combination with the necessary specified functions needs to be designed.

In this work, we report a molecular complementary passivation (MCP) strategy by employing propylphosphonic acid 3-ammonium bromide (PPAABr) in collaboration with phenethylammonium bromide (PEABr) to mutually passivate surface defects of I and formamidinium (FA) vacancies by multi-coordination and pinning the surface Fermi level, which is projected to reduce interfacial non-irradiative recombination and enhance electron transport at that interface. Relying on the chemical interactions between perovskite and PPAABr, MCP surface treatment leads to an obvious defect-density decrease and a more n-type perovskite surface, as demonstrated by a series of comprehensive spectroscopy characterizations. As a result, MCP devices achieved a record PCE of 26.40% (25.92% certified) with a significantly improved fill factor (FF) of 86.3% and an open-circuit voltage (V_{OC}) loss of 362 mV. Relying on this surface treatment, the encapsulated MCP device retains 94.8% of its initial efficiency after 1,000 h of light soaking. Moreover, the MCP strategy also resulted in a competitive PCE of 23.66% for 1.68 eV wide-band-gap (WBG) PSCs, evidencing its generality.

RESULTS

Chemical interactions between PPAABr and perovskite

PPAABr molecule was synthesized by reacting 3-aminopropylphosphonic acid with hydrobromic acid in a molar ratio of 1:2, with the detailed process described in the synthesis method of PPAABr. As Figure S1 shows, its chemical structure was determined by ^{1}H and ^{13}C NMR (Figure S1), along with the mass spectrum (Figure S2). PPAABr contains two critical groups, R-NH^{3+} and R-$PO(OH)_2$, which are projected in the MCP strategy to play such critical roles as filling FA and I vacancies at the interface between perovskite and $PC_{61}BM$. To reveal these roles, we performed density functional theory (DFT) calculations, choosing $FAPbI_3$ as the lattice and constructing an formamidine iodide (FAI)-terminated (001) perovskite surface with two dominant defect states of iodine vacancy (V_I) and FA vacancy (V_{FA}).[24,25] As the calculated results show, for V_I defects, an obvious negative interaction energy (E_{int}) of −3.33 eV is observed when PPAABr is placed and relaxed on the perovskite surface (Figure 1A), revealing that –P=O of PPAABr is strongly inclined to form a dative bond with the exposed Pb at the perovskite surface due to V_I defects. In contrast, the E_{int} between –P–O(H) and the exposed Pb is only −0.10 eV and does not converge (Figures 1B and 1C). Correspondingly, the converged bonding of –P=O—Pb bonds is 2.35 Å and much shorter than that of R-P-O(H) (3.52 Å) after thermodynamic stabilization (Figures 1A–1C), which may be ascribed to the larger electronegativity of –P=O compared with –P–O(H) in propylphosphonic acid 3-ammonium ($PPAA^+$). For V_{FA} defects, as Figures 1D and S3 show, the large negative E_{int} of $PPAA^+$ (−5.17 eV) demonstrates that $PPAA^+$ can spontaneously fill the FA^+ vacancy at the perovskite surface, similar to PEA^+ with a E_{int} of −5.16 eV. This is because they own the identical –NH_3^+ group as FA^+. In fact, when we increased the concentration of PPAABr to 5 mg/mL treatment at the surface of the perovskite film (Figure 1E), a 2D perovskite surface layer involved with $PPAA^+$ was clearly detected by grazing-incident X-ray diffraction

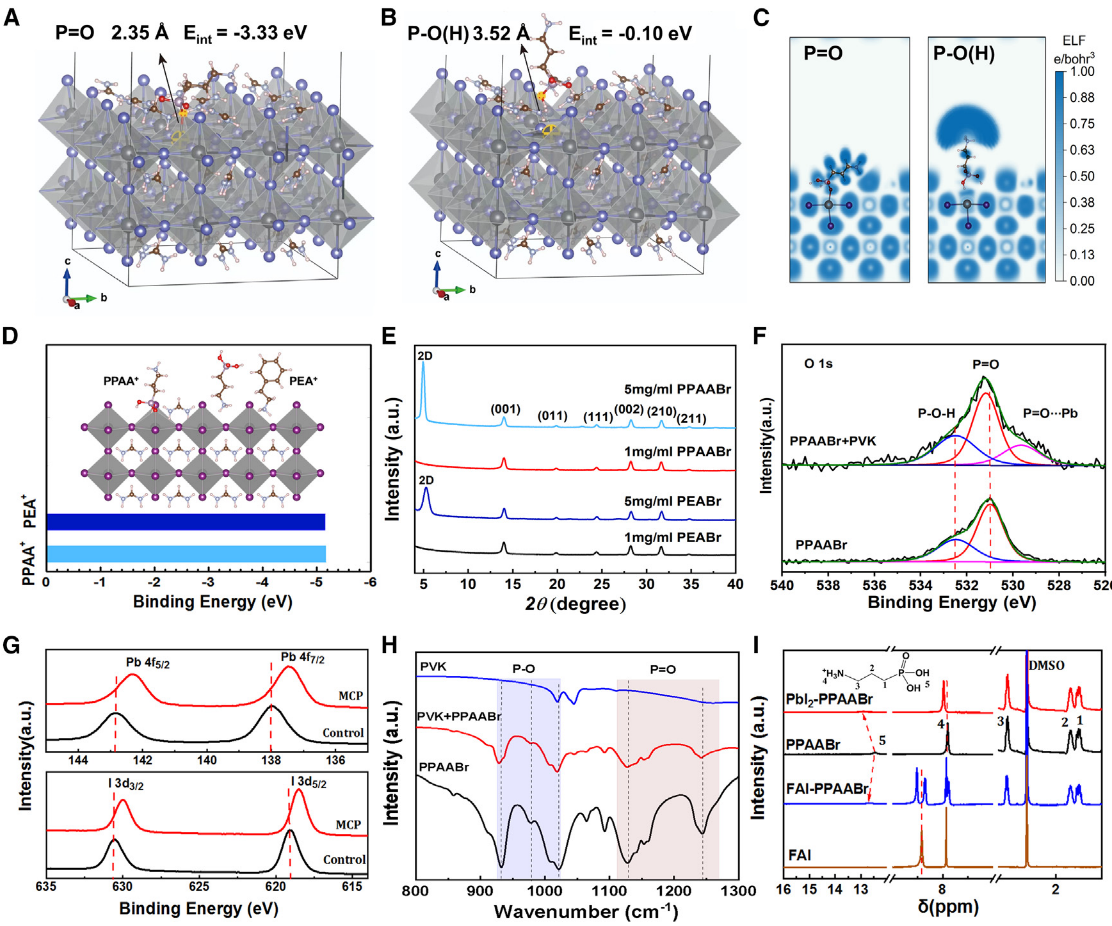

Figure 1. Chemical interactions between perovskite and molecule

(A and B) Optimized structure of $PPAA^+$ with (A) –P=O and (B) –P–O(H) adsorption on FAI-terminated perovskite (001) surface containing V_I defects. E_{int} denotes the interaction energy.

(C) Calculated electron localization function.

(D) Interaction diagram between $PPAA^+$/PEA^+ and V_{FA}/V_I defects at the perovskite surface, and the binding energy (E_b) between $PPAA^+$/PEA^+ and V_{FA} defects.

(E) GIXRD patterns of the perovskite films surfaces treated by PEABr and PPAABr.

(F) XPS spectra of PPAABr with and without perovskite underlayer, deposited on ITO glass, corresponding to the core levels of O 1s.

(G) XPS spectra of control and MCP perovskite flms deposited on ITO glass corresponding to the core levels of Pb 4f (top) and I 3d (bottom).

(H) FTIR spectra of PPAABr, perovskite, and their mixture.

(I) ^1H NMR spectra of PPAABr and its mixture with FAI and PbI_2 in DMSO-d_6 solution.

(GIXRD) with a parallel-beam geometry, similar to PEA^+ that allows $PPAA^+$ ions to not only have a strong propensity to fill the FA^+ vacancy as PEA^+, but also the ability to fill V_I defect states, as schematized in Figure 1D. Moreover, the calculated results indicate that Br^- ions can fill V_I at the $FAPbI_3$ surface with stronger binding energy (E_b) than I^- ions (Figure S4). This suggests that bromide salt molecules have the advantage of eliminating V_I defects at the perovskite surface.[26]

The calculated chemical interactions between PPAABr and perovskite surface defect states were further examined by X-ray photoelectron spectroscopy (XPS), Fourier-transform infrared (FTIR) spectroscopy, and ^1H NMR. Figure 1F shows the O 1s XPS spectrum evolution of PPAABr before and after interaction with the perovskite. The spectra of the pristine PPAABr powder have two peaks around 532.5 and 531.2 eV, indexed to –P–O–H and –P=O, respectively. When PPAABr was spin coated onto the perovskite film, a new peak formed around 529 eV, along with an obvious blue shift of the main-strain –P=O, which can be attributed to the formation of a –P=O—Pb dative bond and is also proved by previous reports.[21,27] In contrast, the peak of –P–OH remains unchanged, which examines again that it is the –P=O group that interacts with the exposed Pb at the perovskite surface, agreeing well with the DFT calculation results (Figures 1A–1C). The obvious E_b redshift of Pb 4f also proves the formation of –P=O—Pb dative bond (Figure 1G). As shown in Figure 1H, the feature stretching vibration frequencies of –P=O of PPAABr (1,229 and 1,244 cm^{-1}) redshift by 2 cm^{-1} when it was blended by milling with our perovskite crystalline powder scraped from the as-deposited perovskite film. This agrees well with the conclusion of dative bond formation between –P=O of PPAABr and Pb of perovskite.[28,29] On the other hand, the two feature stretching vibration frequencies of R-P-OH in PPAABr shift obviously from 932 to 928 cm^{-1} and from 1,021 to 1,019 cm^{-1}, which indicates the formation of –P–O–H•••I hydrogen bonding.[28,30] Further relying on ^{1}H NMR (Figure 1I, red), we observed two obvious shifts when PPAABr was mixed with lead iodide (PbI_2) in dimethyl sulfoxide (DMSO)-d_6 solvent: H of R-$PO(OH)_2$ (no. 5) and H of –NH_3^+ (no. 4). After introducing PbI_2, the downfield shifts from 12.46 to 12.89 ppm for H of R-$PO(OH)_2$ (no. 5) and from 7.81 to 7.96 ppm for H of –NH_3^+ (no. 4) confirm the formation of hydrogen bonding –O–H•••I and –N–H•••I by alleviating the electron cloud density of –O–H and –N–H and deshielding the magnetic nucleus.[30,31] Regarding the interaction between PPAABr and FAI in DMSO-d_6 solution (Figure 1I, blue), the amine H peak of FAI at 8.83 ppm was split into two parts, 9.03 and 8.72 ppm. The peak at 8.83 ppm in the 1H NMR of FAI corresponds to the amine group of FA^+ (Figure 1I, brown), where the C=N bond was found to oscillate between both sides of FA^+ and shows one identical peak in 1H NMR of FAI.[32] This splitting indicates that the formation of hydrogen bonding with R-P-O(H) stabilizes the oscillating H^+ at one end of FA^+ and lets the other end with two nonionic H atoms.[33] The hydrogen bonding between I^- and $PPAA^+$ was also examined by the redshift of the I 3d peaks in XPS spectra (Figure 1G). The above comprehensive spectroscopy characterizations prove well the chemical interactions between $PPAA^+$ and perovskite defects, deepening our understanding of the mechanism for suppressing perovskite surface defect states.

Optoelectronic properties of perovskite films enhanced by MCP treatment

The perovskite involved in this work is $(FA_{0.95}Cs_{0.05}PbI_3)_{0.96}(MAPbBr_3)_{0.04}$ with a band gap of 1.53 eV (Figure S5), doped with DMAcPA.[12] A trace of Cl was detected in the final perovskite film when methylamine hydrochloride (MACl) was used to enhance the film crystallinity (Figure S6).[34] With MCP surface treatment, the absorption edge remained unchanged while the absorption intensity was enhanced, as observed in the UV-vis absorption spectra (Figure S5). The latter indicates that its optoelectronic property improved by MCP treatment. MCP-treated perovskite film also shows a negligible change in its X-ray diffraction (XRD) patterns (Figure S7). To investigate the effect of surface molecular coordination discussed above, we conducted systematic spectroscopy characterizations of the perovskite films. When the perovskite film was irradiated with a 375 nm laser from the top surface (see the inset in Figure 2A), time-resolved photoluminescence (TRPL) spectra revealed a remarkable difference of exciton lifetime between the perovskite film without (control) and with MCP surface treatment (MCP) (Figure 2A), increasing from 1,608.26 to 3,952.18 ns after the surface treatment (Table S1). The steady-state photoluminescence (PL) intensity augment also proves the effect of MCP surface treatment (Figure S8A). They both clearly demonstrate that the perovskite surface defect states were well passivated by PPAABr.[35,36] In comparison, if we irradiated the perovskite film at the glass side, both TRPL and PL spectra showed negligible variation (Figures 2B and S8B; Table S2)s. This result is in accordance with the fact that the buried interfaces of both control and MCP are in the same structure.[12] Transient absorption (TA) measurements provide us with further clues about the recombination pathway to further investigate charge carrier dynamics. Considering the carrier recombination process caused by surface defect states in perovskite films, a three-species sequential model is used in TA for the global analysis, including hot carrier cooling (SA1), energy transfer (SA2), and charge recombination (SA3).[37] As illustrated in Figures 2C–2F, the ground bleach was featured at 763 nm (Figures 2C and 2D), and the hot carrier was relaxed at around 0.3 ps. MCP showed a longer charge recombination lifetime of 14.6 ns compared to control (10.5 ns), further demonstrating less interfacial recombination of MCP (Figures 2D and 2F). PL mappings shown in Figures 2G and 2H further testify the suppressing effect of surface defect states that much more uniform and higher PL intensity was obtained by MCP treatment at the perovskite surface. For the whole device, the trap density of states (tDOSs) was measured by admittance spectroscopy (Figure 2I). The deep-level defect density of MCP devices is obviously lower than that of the control device across the energy span from 250 to 500 meV, demonstrating the reduced density of deep-level defect states owing to the passivation effect of $PPAA^+$, which is consistent with the above discussion. These above analyses exhibit well that our MCP surface treatment effectively diminishes the deep-level surface defects of perovskite films and improves the film's optoelectronic properties.

Interface energy level alignment modulated by MCP treatment

Besides defect passivation, the top surface work function (W_F) of the perovskite film needs to be investigated, which decides the electron transfer at that interface.[23] Figures 3A and 3D show atomic force microscopy (AFM) images of the perovskite film surface morphologies before and after MCP treatment. The roughness decreases moderately from 14.8 to 11.9 nm, which correlates with the comparison of their corresponding scanning electron microscopy (SEM) top-view images (Figure S9). A smoother perovskite surface is beneficial to form a continuous ETL thin layer and thus a high-quality interface.[38,39] Depending on Kelvin probe force microscopy (KPFM), their surface potential imaging was recorded (Figures 3B, 3E, and S10). The difference in surface roughness between control and MCP is very small, except for the sutures of grain boundaries (Figures 3A and 3D). This suture can be verified more obviously by the surface potential line profiles shown in Figures 3C and 3F. The surface potential variation shrinks remarkably after PPAABr surface modification, with the contact potential difference (ΔV_{CPD}) decreasing from 33 to 25 mV, which is attributed to the filling of PPAABr into grain boundaries that flattens the potential contributed by PPAABr molecules.[15,40] A smaller potential difference means that the capture of charge carriers at grain boundaries can be reduced and favored promoting charge extraction, thereby reducing the leakage current of the device and improving its FF and V_{OC}. After MCP surface treatment, the mean surface potential of the perovskite film deviated significantly from -65 to -532 mV, compared with the Au reference, which declares that the W_F of the perovskite top surface was shifted to a shallower level.[41] That is also confirmed by the

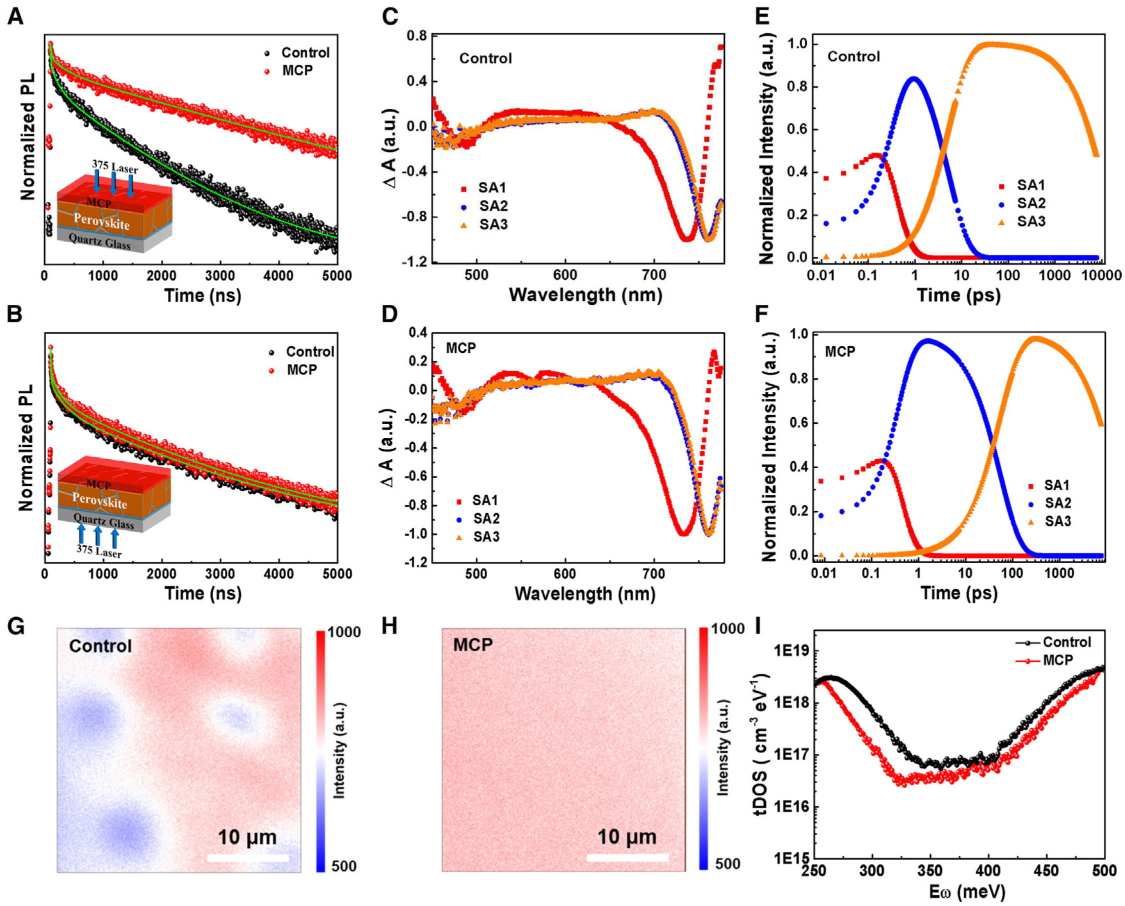

Figure 2. Optoelectronic properties and defect density characterizations

(A and B) TRPL illuminated at the (A) top surfaces and (B) bottom buried interfaces of control and MCP perovskite films deposited on quartz glass.
(C and D) Deconvoluted species associated TA spectra using 400 nm excitation of (C) control perovskite film and (D) MCP perovskite film.
(E and F) Population dynamics of (E) control perovskite film and (F) MCP perovskite film.
(G and H) Two-dimensional PL intensity of (G) control and (H) MCP samples.
(I) tDOS spectra for control and MCP PSC devices.

characterization results of ultraviolet photoelectron spectroscopy (UPS). The W_F of the perovskite film top surface turned shallower from 4.62 to 4.37 eV (Figures 3G and 3I).[42,43] According to the Tauc plot of the absorption spectra (Figure 3H), the optical band gaps of control and MCP perovskite films display the same band edge and can be determined to be 1.53 eV identically. Correlated with the cutoff values in the left graph of Figure 3G, we can determine the valence band maximum (VBM) levels at −5.54 and −5.56 eV for control and MCP, respectively. This indicates that MCP surface treatment does not change any structure of the perovskite film but remarkably tailors the perovskite surface Fermi level (E_F) to be n-type and closer to that of ETL ($PC_{61}BM$). TRPL and PL spectra in Figure S11 verify that MCP has more efficient interfacial charge transport than control in the same device structure as our solar cell (Table S3). This augments the built-in field within the perovskite film for more efficient electron-hole dissociation and also favors electron extraction at the perovskite/$PC_{61}BM$ interface.[44,45]

Device performance and device physics analysis

With the effective defect-passivation effect of MCP surface treatment, a device with a structure of ITO/perovskite (DMAcPA) with or without MCP/$PC_{61}BM$/bathocuproine (BCP)/Ag was fabricated to investigate the photovoltaic performance (the inset in Figure 4A). As Figure 4A shows, the MCP device got a champion PCE of 26.40% with an V_{OC} of 1.168 V, a short-circuit current density (J_{SC}) of 26.19 mA cm^{-2}, and an FF of 86.3%, compared with an optimal PCE of 25.12% for control. The corresponding *J-V* curves in both forward and reverse scan directions for the champion devices of control and MCP are plotted in Figure S12 for comparison. The MCP device exhibits a much lower hysteresis of 0.64% compared to 2.59% for the control (Table S4), which declares that MCP treatment eliminates the hysteresis obviously by enhancing the interfacial electron transfer. For comparison, with only PPAABr in the perovskite film surface treatment, 25.53% of device PCE was obtained (Figure S13), superior to PEABr-only or control. This can be attributed to $PPAA^+$ passivating both V_{FA}

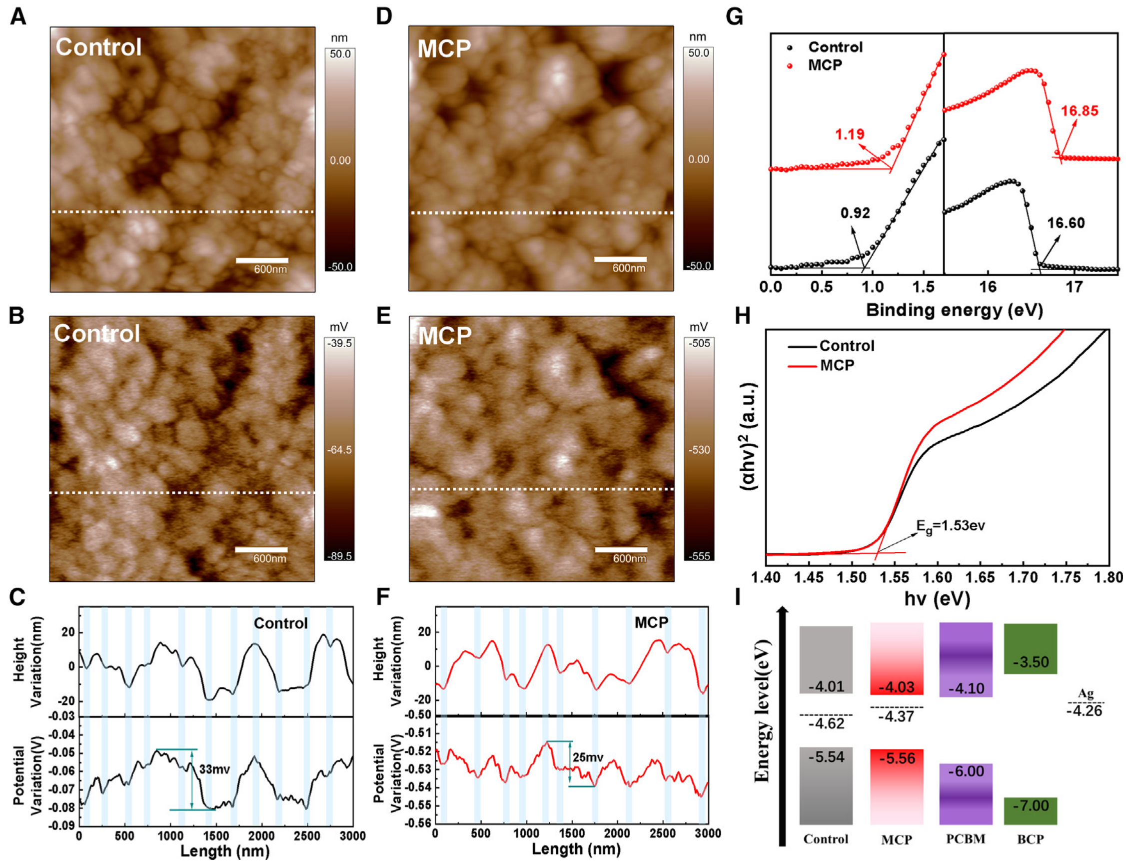

Figure 3. Surface morphology and potentials of perovskite film before and after MCP treatment

(A–F) (A and D) Surface morphology, (B and E) surface potential images, and (C and F) line profiles of control and MCP perovskite films by KPFM. The four lines are marked in (A), (B), (D), and €.

(G) Onset (left) and cutoff regions of top surfaces of control and MCP perovskite films, characterized by UPS.

(H) Tauc plots of UV-vis absorption spectra of control and MCP perovskite films.

(I) Energy level diagram of the perovskite/PCBM interface extracted from (G).

and V_I surface defect states, while PEA^+ may only fill FA vacancies (Figure 1D), examining the significance of MCP. Instead of PPAABr, the device performance with PPAAI is slightly inferior to that of PPAABr in our MCP surface treatment (Figure S14), which is ascribed to the stronger E_b of bromide ions interacting with the surface V_I defects (Figure S4). Steady power output (SPO) of both control and MCP devices in Figure 4B proves the reliability of our *J-V* testing. The PCE of the MCP device was certified to be 25.92% by the Shanghai Institute of Microsystems and Information Technology (Figure S15). In the branch of PSCs without predeposited HTLs, our MCP surface treatment advances device performance to a new level (Figure S16). As Figure 4C shows, the J_{SC} integrated from the external quantum efficiency (EQE) spectrum agrees well with that obtained from *J-V* curves. The statistics on a batch of 35 cells prepared under the same experimental conditions demonstrate good reproducibility of our fabrication process (Figures 4D and S17). The average device PCE was improved from 24.40% to 25.91%, along with increments of V_{OC} from 1.141 to 1.164 V, J_{SC} from 25.86 to 26.16 mA cm^{-2}, and FF from 0.827 to 0.851. As discussed above, MCP surface treatment contributes to the remarkable enhancement of device photovoltaic performance.

To demonstrate our MCP process scalability, an MCP device with an active area of 1 cm^2 exhibits a competitive PCE of 24.87% among the cutting-edge reported values (Figure 4E),[2,46] compared with 23.58% for the control. To validate the generality of the MCP strategy, we fabricated WBG PSCs with a 1.68 eV $Cs_{0.05}FA_{0.8}MA_{0.15}PbI_{2.25}Br_{0.75}$ as the absorber,[47] resulting in an outstanding PCE of 23.66% (Figures 4F and S18; Table S5). Moreover, the MCP device also shows an outstanding stability of maintaining 94.8% of its initial PCE after 1,000 h of light soaking under maximum power point (MPP) tracking mode by using a light-emitting diode lamp with a calibrated light intensity of 100 mW/cm^2 (Figure 4G). In contrast, the control device degraded faster, retaining 85.1% of the initial efficiency. As the damp-heat stability test results show (Figure 4H), the MCP device kept 92.4% of the initial conversion efficiency after a duration of 500 h, compared with 85.6% for the control. Both devices exhibit good damp-heat stability.

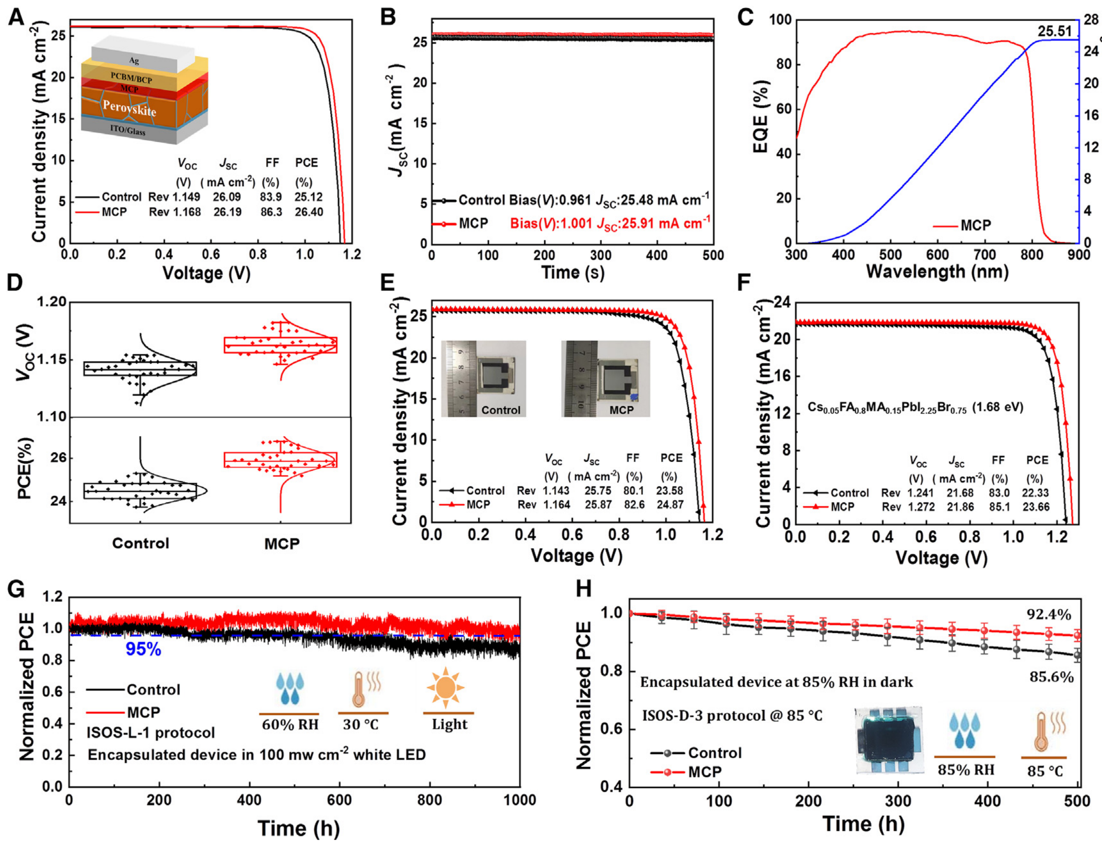

Figure 4. Device performances

(A and B) (A) *J-V* curves and (B) steady power output (SPO) of control and MCP champion devices.

(C) EQE spectra and integrated J_{SC} of MCP champion device.

(D) Statistical distribution of V_{OC} (top) and PCE (bottom) for control and MCP devices, collected from a batch of 35 cells.

(E) *J-V* curve of control and MCP device with 1 cm^2 active area.

(F) *J-V* curves of $Cs_{0.05}FA_{0.8}MA_{0.15}PbI_{2.25}Br_{0.75}$ based control and MCP champion devices.

(G) Normalized PCE of the encapsulated PSCs of control and MCP champion devices tested in MPP tracking mode under a continuous 100 mW/cm^2 illumination in ambient environment.

(H) Damp-heat reliability tests for the encapsulated devices tested at 85°C and 85% relative humidity. The error bars for MCP and control are 2.1% and 2.4%, respectively.

Along with the MPPT results, we can attribute the strong stability to the well-round grain boundary passivation by DMAcPA molecules. The improvement of the MCP device can be ascribed to PPAABr passivating interface iodine vacancies, while phosphate groups can form hydrogen bonds with iodine to enhance the barrier of iodine migration.[30] This suppresses iodine migration to the rear Ag electrode for the irreversible degradation.[23]

Depending on the complementary passivation effect of PPAABr and PEABr, we carried out comprehensive device physics analysis to further reveal the improvement relationship between device performance and defect states. In Figure 5A, the MCP device shows a transient photovoltage (TPV) decay much slower than the control, which examines again the reduced defect density by MCP surface treatment. Meanwhile, a faster transient photocurrent (TPC) decay of the MCP device declares that MCP has a faster charge extraction rate, compared with control (Figure 5B). Both would contribute to the augment of V_{OC}.[48,49] That was clearly verified by the light intensity-dependent V_{OC} plot (Figure 5C). The control device has a slope of 1.68 k_BT/q, while MCP has a much smaller slope of 1.24 k_BT/q, where k_B is the Boltzmann constant, T is the temperature, and q is the elementary charge. A slope closer to 1 means that MCP has a lower defect density.[50] According to the results of electrochemical impedance spectroscopy (EIS), the recombination resistance (R_{rec}) increased from 22.7 to 58.3 kΩ after MCP treatment (Figure 5D), which demonstrates suppressed trap-assisted recombination and would improve both V_{OC} and FF.[51] Moreover, Figure S19 shows that MCP leads to a lower leakage current owing to the decrease of trap states. Furthermore, based on space charge-limited current (SCLC) measurement, the trap-state density of perovskite in the hole-only device can be determined. By linearly fitting the three SCLC regions in Figure 5E, the trap-filling limit voltage (V_{TFL}) of control and MCP devices is deduced to be 0.13 and 0.08 V, respectively. Following the equation N_t =

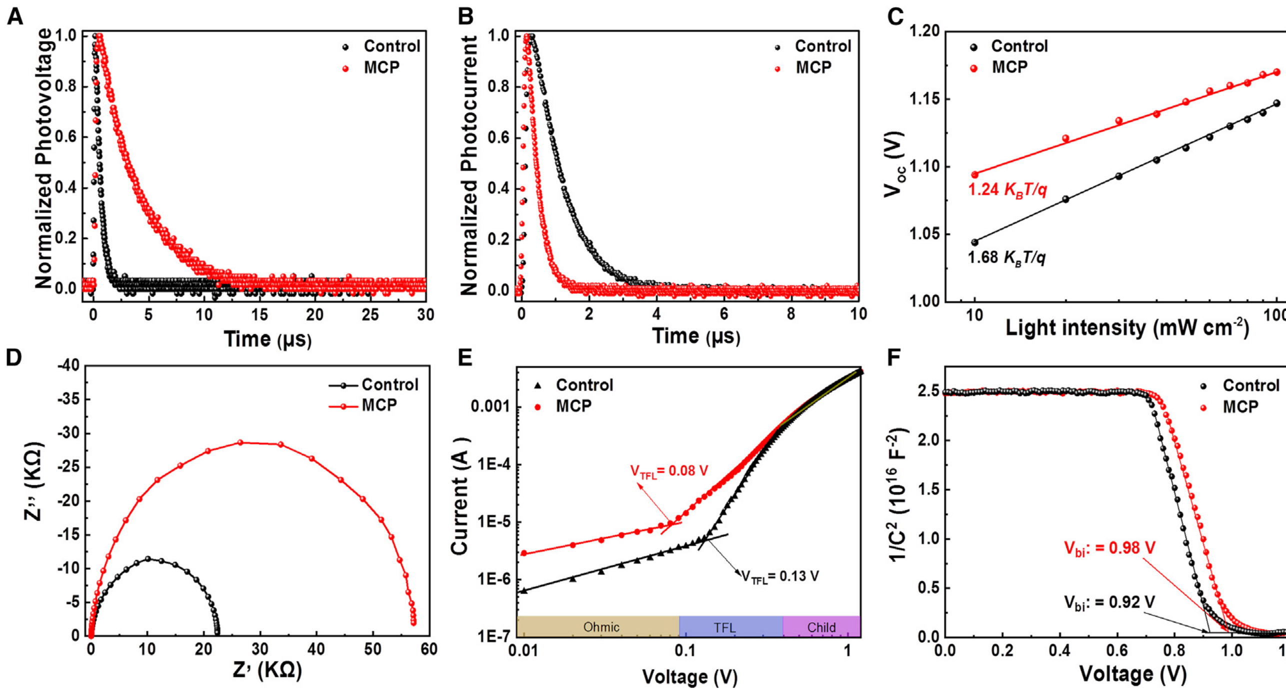

Figure 5. Device physics analysis

(A–D) (A) TPV, (B) TPC, (C) V_{OC} versus light intensity, and (D) Nyquist plots of control and MCP devices.

(E) SCLC measurements of electron-only devices based on control and MCP samples.

(F) Mott-Schottky plots of control and MCP devices.

$2\varepsilon\varepsilon_0 V_{TFL}/eL^2$,[39] where ε_0 is the vacuum permittivity, ε is the relative dielectric constant, e is the elementary charge, and L is the thickness of the perovskite layer, the trap density (N_t) is calculated as 1.05×10^{15} cm^{-3} for control and 0.65×10^{15} cm^{-3} for MCP. That clearly demonstrates the defect density variation after MCP treatment. Finally, Mott-Schottky (M-S) plots in Figure 5F disclose that the built-in field (V_{bi}) was enhanced from 0.92 to 0.98 V after MCP treatment, which means an enhanced driving force for the dissociation and transport of photo-generated carriers.[52] That enhancement is consistent with the V_{OC} augment tested in *J-V* curves (Figure 4A).

DISCUSSION

In summary, we employed PPAABr salt molecules with PEABr to successfully address the defect-passivation issue of perovskite/ETL interface in HTL-free PSCs, called the MCP strategy. Through comprehensive characterizations such as FTIR, ^{1}H NMR, and XPS, along with DFT calculations, PPAABr was testified to passivate V_I and V_{FA} surface defects of perovskite film by phosphonic acid and amine groups. This leads to an obvious decrease of interfacial defect-state density and a much-improved exciton and carrier lifetime for the perovskite film. Moreover, MCP surface treatment also pushes the Fermi level of the perovskite surface layer closer to that of $PC_{61}BM$ ETL and thereby enhances the interfacial electron extraction. As a result, the MCP-based HTL-free PSC achieved a champion efficiency of 26.40%, along with a certified efficiency of 25.92%. Relying on the MCP surface treatment, the encapsulated MCP device retains 94.8% of its initial efficiency after 1,000 h of light soaking. Systematic device physics analysis was conducted to reveal the positive relationship between material properties and device performance enhancement. Our MCP strategy takes a good generality to obtain a competitive PCE of 23.66% for 1.68 eV WBG PSCs and paves a new way to address the critical interfacial issues and promote the further progress of high-performance PSCs.

METHODS

Materials

PbI_2 (99.99%) was purchased from Tokyo Chemical Industry. Cesium iodide (CsI, 99.999%) was purchased from Sigma-Aldrich. MACl (99.9%) and FAI (99.99%) were purchased from Great Cell Solar. BCP and PEABr were obtained from Xi'an Polymer Light Technology. Hydrogen bromide, magnesium fluoride (MgF_2), and 3-aminopropylphosphonic acid were purchased from Aladdin. DMSO-d_6 was purchased from the Macklin. Chlorobenzene (CB), DMSO, N,N-dimethylformamide (DMF), and isopropanol (IPA) were purchased from Sigma-Aldrich. ITO glass was purchased from Suzhou ShangYang Solar Technology. Methylammonium lead bromide ($MAPbBr_3$) was purchased from Opvtech. $PC_{61}BM$ was purchased from Lumtec. DMAcPA was synthesized in our laboratory, and the detailed synthesis process can be found in previous articles.[12]

Synthesis method of PPAABr

3-Aminopropylphosphonic acid was reacted with excess hydrobromic acid in a molar ratio of 1:2 and stirred in an ice water bath for 1 h. Then, the remaining liquid was removed by rotary evaporation under reduced pressure at 65°C. The precipitate was washed three times with ether and vacuum dried overnight at 60°C.

PSC fabrication

ITO/glass substrates (25 × 25 mm^2) were meticulously cleaned through a sequence of ultrasonic treatments involving detergent, acetone, and IPA. After this cleaning regimen, the substrates were carefully dried using nitrogen gas before being subjected to ultraviolet (UV) light for a period of 60 min.

Normal band-gap perovskite

A perovskite precursor solution with a molar concentration of 1.5 M was prepared using a stoichiometric composition of $FA_{0.95}Cs_{0.05}PbI_3$ by combining the appropriate amounts of FAI, PbI_2, and CsI in 1 mL of a mixed DMF/DMSO (4:1) solvent system. Additionally, 4 mg of DMAcPA powder, 4 mol % $MAPbBr_3$, and 12.5 mol % MACl were added to enhance crystallinity. The mixture was then stirred at 50°C for 30 min to ensure thorough mixing and dissolution.

1.68-eV WBG perovskite

$Cs_{0.05}MA_{0.15}FA_{0.8}PbI_{2.25}Br_{0.75}$ perovskite precursor solution was prepared by mixing 1.04 mM FAI, 0.195 mM MABr, 0.065 mM CsI, 0.39 mM $PbBr_2$, and 0.91 mM PbI_2 in 1 mL of a mixed DMF/DMSO (4:1) solvent system. An additional 4 mg of DMAcPA and 7 mol % MACl powder were added to the perovskite solution. The mixture was then stirred at 50°C for 30 min to ensure thorough mixing and dissolution.

The inverted PSC studied in this paper with an architecture of ITO/perovskite/$PC_{61}BM$/BCP/Ag. 80 μL of perovskite precursor solution was dripped onto the ITO/glass substrates and spin coated at 1,000 rpm for 10 s and 4,000 rpm for 30 s, then antisolvent CB was dripped dropwise after the penultimate 5 s. Thereafter, the perovskite films were immediately annealed at 105°C for 30 min.

For the preparation of the control perovskite layer, PEABr with a concentration of 1 mg ml^{-1} in a mixed IPA/DMSO (volume ratio 200:1) solution was spin coated on the perovskite layer substrates at 4,500 rpm for 30 s, then annealing at 100°C for 5 min.

For the preparation of the MCP perovskite layer, a mixture of PEABr (1 mg mL^{-1}) and PPAABr (0.3 mg mL^{-1}) was dissolved in a mixed solvent of IPA:DMSO (200:1 in volume) and dynamically spun on the as-prepared perovskite films at 4,500 rpm for 30 s, then the perovskite films were immediately annealed at 100°C for 5 min. Subsequently, $PC_{61}BM$ (20 mg mL^{-1} in CB, 1,500 rpm, 35 s) was coated and annealed at 100°C for 10 min, followed by spin-coating BCP (0.5 mg ml^{-1} in IPA) on top (4,500 rpm, 30 s). The sample was deposited with a 100 nm Ag layer in a hot evaporator to complete the device. Finally, a 120 nm thick MgF_2 layer was deposited on the back of the ITO substrate to enhance transmittance.

Device characterization

The NMR spectra were recorded on a Bruker Advance 500 spectrometer in DMSO-d_6. High-resolution mass spectra (HRMS) were recorded on a Thermo Scientific Q Exactive mass spectrometer. FTIR spectra were recorded using a Bruker Vertex v70 instrument. The XRD spectra were recorded using a Bruker D8 ADVANCE, at a scan rate of 6° min^{-1} and a 2θ range of 5°–45°. The absorption spectra were recorded on a Shimadzu spectrophotometer. The surface roughness and potential were tested using an MFP-3D-Stand Alone Atomic Force Microscope (AFM, Oxford Instrument). SEM was characterized using a HITACHI SU8230. XPS and UPS measurements were carried out on an Omicron ESCA Probe spectrometer (Thermo Scientific ESCALAB 250Xi). Steady-state PL and TRPL spectroscopy were measured using an Edinburgh FLS 980 spectrometer. The current density-voltage (*J-V*) measurements were carried out using a Keithley 2400 source meter in an ambient environment under standard AM 1.5 G illumination (Enli Technology, Taiwan). The unencapsulated devices were performed in both forward (from −0.02 to 1.2 V) and reverse scanning (from 1.2 to −0.02 V) with a delay time of 100 ms. The effective area of the device (0.08 cm^2) was calibrated using a shadow mask during the measurements. EQE spectra were measured with a QE-R 3011 EQE system (Enli Technology, Taiwan) ranging from 300 to 900 nm. SCLC measurements were measured using a Keithley 4200. Impedance spectroscopy and Mott-Schottky analyses were conducted on the Zahner Zennium electrochemical workstation. TPV and TPC measurements were carried out using a 405 nm pulsed laser and oscilloscope (TDS 2024C). TA spectra of perovskite films were measured using an Ultrafast System HELIOS TA spectrometer.

DFT calculation method

All the DFT calculations were implemented by using the Vienna Ab initio simulation package (VASP).[53] The exchange-correlation potential was described through the generalized gradient approximation of the Perdew-Burke-Ernzerhof (GGA-PBE) method. The interactions between ion cores and valence electrons were treated by the projector-augmented wave (PAW) method. The plane-wave cutoff energy was set to 520 eV. A Γ-only k-mesh was used for these calculations. All these structures were fully relaxed until the Hellmann-Feynman forces were smaller than 0.02 eV/Å, and the energy between two electronic steps was smaller than 10^{-5} eV. The vdW interactions were corrected by the DFT + D3 method.[54]

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Zhubing He (hezb@sustech.edu.cn).

Materials availability

This study did not generate new unique materials

Data and code availability

This study did not generate or analyze datasets or code.

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AUTHOR CONTRIBUTIONS

Z.H. and Q.C. conceived the idea for the project. Z.H. supervised the project. Q.C. fabricated and characterized the devices and related films. Q.T. and S.T. taught Q.C. the molecule-extrusion process. Q.S. conducted IR characterization. D.H. performed EIS and M-S testing. T.C. performed XPS and UPS characterization. J. Huang conducted NMR characterization. G.M. conducted TPV and TPC characterization. C.C. and G.S. conducted AFM characterization. J.F. conducted DFT calculations. H.G., B.W., and G.X. performed and analyzed TA characterization. Q.C. and J. He analyzed all data and wrote the manuscript. Z.H. revised the manuscript. All authors discussed the results and commented on the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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